# HALO

## HIGH ACUITY LOW OCCURRENCE EMERGENCY PROCEDURES

**🩺 PAEDIATRIC EDITION — PATIENTS UNDER 16 YEARS**

**FOR PATIENTS UNDER 16 YEARS OF AGE**
DO NOT USE THE ADULT EDITION FOR PAEDIATRIC EMERGENCIES

**2026 PAEDIATRIC EDITION**

Version 1.3 (Sep 2026)

13 Procedures | 31 Pages | All Ages incl. Neonates

WETFLAG Reference | DAS 2025 | RCUK PALS 2025

KIDS NTS: 0300 200 1100

Based on RCOG GTG-56 (2019, reviewed Dec 2024), Resuscitation Council UK (2025 Guidelines), DAS 2025 | Review due Sep 2027

# HOW TO USE THIS GUIDE

## 1. The Two-Page System

**Page A: Walkthrough (Preparation)** — Use during the initial "Kit Dump" phase. Contains equipment checklists, team roles, and setup instructions.
**Page B: Flowchart (Action)** — Use during the procedure. Designed for high-stress cognitive processing with critical steps highlighted in red.

## 2. Design Principles

**Standard Steps (Grey):** Routine actions and setups.
**Critical Steps (Red):** Dangerous/irreversible actions — pause and confirm.
**Paeds Tip (Green):** Paediatric-specific modifications.
**Pro-Tip (Blue):** Technique pearls and cognitive aids.
**Plan B (Red Box):** Fallback if primary method fails.

## 3. Human Factors & Safety

**Allocate a Reader:** Hand this booklet to a specific team member to read steps aloud, reducing cognitive load for the operator.
**Pre-Procedure Pause:** 5-second pause before any incision. Confirm landmarks verbally.
**Weight Estimation:** Use Broselow tape if available. Formula: <1yr=(0.5×months)+4, 1-5yr=(2×age)+8, 6-12yr=(3×age)+7.

## 4. Glossary of Abbreviations

| | | | |
|---|---|---|---|
| **DAS** | Difficult Airway Society | **RCOG** | Royal College of Obstetricians & Gynaecologists |
| **RCUK** | Resuscitation Council UK | **RCPCH** | Royal College of Paediatrics & Child Health |
| **NLS** | Neonatal Life Support | **PALS** | Paediatric Advanced Life Support |
| **ROSC** | Return of Spontaneous Circulation | **LUD** | Left Uterine Displacement |
| **ETT** | Endotracheal Tube | **TXA** | Tranexamic Acid |
| **GTG** | RCOG Green-top Guideline | **NICU** | Neonatal Intensive Care Unit |
| **PMCS** | Perimortem Caesarean Section | **RH** | Resuscitative Hysterotomy |
| **ACC** | Aortocaval Compression | **IO** | Intraosseous |
| **TBSA** | Total Body Surface Area | **CICO** | Can't Intubate Can't Oxygenate |
| **CTM** | Cricothyroid Membrane | **eFONA** | Emergency Front of Neck Airway |
| **LAST** | Local Anaesthetic Systemic Toxicity | | |

⚠️ **CLINICAL DISCLAIMER**
These guides are cognitive aids for trained medical professionals and do not replace clinical judgement. Guidelines are subject to change. For up-to-date guidance, consult RCOG GTG-56 (Dec 2024), RCUK 2025, DAS 2025, and RCPCH guidelines. Weight-based drug dosing is mandatory in all paediatric patients.

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🩺 = Paediatric-specific modification


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## WETFLAG

Weight · Energy · Tube · Fluids · Lorazepam · Adrenaline · Glucose

**W Weight**
<1yr: (0.5×months)+4
1–5yr: (2×age)+8
6–12yr: (3×age)+7 kg

**E Energy (defibrillation)**
**4 J/kg** initial shocks | refractory VF/pVT (i.e. more than five shocks are needed): stepwise up to 8 J/kg (max 360 J)

**T Tube (ETT)**
Uncuffed: (Age÷4)+4 mm
Cuffed: (Age÷4)+3.5 mm
Oral depth: (Age÷2)+12 cm

**F Fluids**
**10 ml/kg** medical | **10 ml/kg** trauma
Hartmann's / Plasmalyte. Reassess after each bolus.

**L Lorazepam**
**0.1 mg/kg** IV/IO | max 4 mg per dose

**A Adrenaline (arrest)**
**0.1 ml/kg of 1:10,000**
= 10 mcg/kg (max 1 mg) | IV/IO | every 3–5 mins

**G Glucose (hypoglycaemia)**
**2 ml/kg of 10% dextrose**
IV/IO. Recheck BM at 5–10 mins.

**Amiodarone:** 5 mg/kg IV/IO after 3rd shock (max 300mg) and 5th shock (max 150mg) | **Atropine:** 20 mcg/kg (max 500mcg up to 11 yrs; 300–600mcg at 12–17 yrs) → See Pg 30 for full WETFLAG reference

## ⚠️ KEY PAEDIATRIC DIFFERENCES AT A GLANCE

**Airway**
ETT cuffed: (Age÷4)+3.5
ETT uncuffed: (Age÷4)+4
eFONA ≥8yrs: scalpel, 6.0 cuffed
<8yrs: surgical tracheotomy
Shiley 3.5 neo / 4.5 paed

**Chest Drain**
Size (Fr) = 4 × ETT size
No trochar in any child
Needle decompression: 14–16G

**Central Line**
IO preferred in arrest

**Pacing**
Rate: 80–100 bpm
Pads <10kg: paediatric

**Cardioversion**
Sync: 1 → 2 → 4 J/kg
Defib: 4 J/kg (refractory: up to 8 J/kg)

**Trauma / Transfer**
MTC: Birmingham Children's
Activate WMAS/MERIT early
KIDS retrieval team

**Burns**
Resus threshold: ≥10% TBSA

📞 **KIDS NTS — 24hr Retrieval & Advice** **0300 200 1100**



Neonates to 16yrs | West Midlands | Advice + Transport | If not connecting: **0121 333 9687**

# RESUSCITATIVE HYSTEROTOMY

Page 1: Walkthrough — PAEDIATRIC (<16 YRS)

## INITIAL ACTIONS & INDICATIONS

- Call **2222** — "Paediatric Obstetric / Cardiac Arrest"
- Activate: EM Reg (lead), Paediatric Reg, Obstetric Reg, Anaesthetics, Neonatal Team (NLS)
- **Indication:** Cardiac arrest AND uterus ≥ umbilicus (≈≥20/40)
- If no ROSC within **4 minutes** of effective CPR → proceed with RH
- Goal: deliver within **5 minutes** of arrest

**ALS Modifications (begin immediately):**

- Manual LUD — push uterus LEFT with both hands
- IV access ×2 above diaphragm (or IO if difficult)
- Early intubation: ETT = `(Age÷4)+3.5` (cuffed), assume difficult airway
- Remove fetal monitors

## EQUIPMENT CHECKLIST

- ☐ Chlorhexidine (rapid splash)
- ☐ **Sterile Drape / Gown Pack**
- ☐ **Scalpel: Size 10 or 20**
- ☐ **Tuff Cut Scissors**
- ☐ **Suction ×2**(Yankauer + wide bore)
- ☐ Large Swabs (for packing)
- ☐ Stapler / Clingfilm / Large Dressings
- ☐ **Umbilical Cord Clamps ×2**
- ☐ **Artery Forceps ×2**
- ☐ Scissors or Scalpel (for cord)
- ☐ **Age-appropriate ETT + BVM**
- ☐ **Broselow Tape**

**📍 STORAGE LOCATION**

**NEONATAL RESUS KIT IN PAEDIATRIC RESUS BAY**

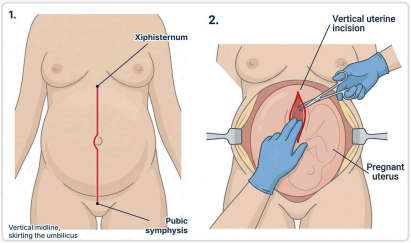
## PROCEDURE WALKTHROUGH

1. **Prepare:** Rapid chlorhexidine splash. Continue CPR throughout.
2. **Skin Incision:** Vertical midline — Xiphisternum to Pubic Symphysis. Deep through skin and fat with SCALPEL.
3. **Peritoneum:** Bluntly enter with fingers (tear open) — avoids bowel injury.
4. **Expose Uterus:** Retract abdominal wall laterally.
5. **Uterine Incision:** Vertical in upper segment of uterus. SCALPEL to start, extend with TUFF CUT SCISSORS.
6. **Protect Baby:** Insert two fingers under uterine wall, away from fetal head. Extend cut superiorly.
7. **Delivery:** Scoop head and shoulders. Do NOT pull by a single limb.
8. **Cord:** Clamp ×2. Cut between clamps. Hand to NLS Team immediately.
9. **Placenta:** One gentle tug. If stuck — leave in situ. Pack uterus with SWABS.

🩺 Adolescent uterus may be smaller — start with smaller stab incision

## POST-PROCEDURE PLAN

**ROSC:** Mass closure (dressings/clingfilm/staples). Emergent Theatre. Continue ALS during transfer.
**Drugs:** Oxytocin `5 units` IV slowly (NEVER rapid bolus — risk of re-arrest), Antibiotics, TXA `15mg/kg IV (max 1g)`
**Neonate:** Dedicated NLS team only. NICU activation ≥22/40.




*Vertical midline incision — speed over sterility*

# RESUSCITATIVE HYSTEROTOMY

Page 2: Action Flowchart — PAEDIATRIC (<16 YRS)

- **GOAL: DELIVERY <5 MINS** | **Indication: Cardiac arrest. Uterus ≥ umbilicus. No ROSC after 4 mins CPR**
- **Call 2222 "Paediatric Obstetric Arrest"**

## 0 IMMEDIATE ALS MODIFICATIONS

- **CPR:** Standard ALS + Manual LUD (push LEFT — do NOT tilt)
- **Airway:** ETT = `(Age÷4)+3.5` cuffed. Assume difficult airway.
- **IV Access:** ×2 above diaphragm. Use IO if peripheral difficult.

Remove fetal monitors — impede resuscitation quality

## 1 ABDOMINAL INCISION

- Pause 5 seconds. Confirm landmarks verbally.
- **Vertical Midline:** Xiphisternum to Pubic Symphysis.
- Deep through skin and fat. Bluntly separate rectus muscles.
- Enter peritoneum bluntly with fingers — do not use blade.

Ignore bleeding — no cardiac output during arrest.

## 2 UTERINE INCISION

- **Vertical incision** upper uterine segment (avoids bladder/arteries).
- Insert two fingers to protect fetal face.
- Extend with TUFF CUT SCISSORS towards fundus.

🩺 Adolescent uterus may be smaller — use smaller initial stab.

## 3 DELIVERY

- Scoop head and shoulders. Do NOT pull on single limb.
- Rotate to disengage from pelvis if needed.
- Clamp cord ×2. Cut between clamps. Hand to NLS team.

## 4 POST-DELIVERY & CLOSURE

- **Placenta:** One gentle tug. If stuck — leave in situ. Pack with SWABS.
- **Drugs:** Oxytocin `5 units IV slowly`, Antibiotics, TXA `15mg/kg (max 1g)` FETCH DRUGS
- **Closure:** Pack uterus. Dressings/clingfilm/staples.
- **Transfer:** Emergent Theatre if ROSC.

⚠️ **PAEDIATRIC NOTE: OXYTOCIN**

Give 5 units IV over ≥1 minute. Never rapid bolus in adolescents — profound hypotension risk. Consider carboprost/ergometrine (obstetric advice).

OBSTETRIC EMERGENCY

*Feedback/Updates: jake.turner@uhb.nhs.uk*

References: RCOG GTG-56 (2nd ed. 2019; reviewed Dec 2024) | RCUK 2025 Special Circumstances

Open RCOG GTG-56

# EMERGENCY BREECH DELIVERY

Page 1: Walkthrough — PAEDIATRIC (<16 YRS)

PAEDIATRIC EDITION

## INITIAL ACTIONS

• Call **2222** "Obstetric Emergency" + "Neonatal Emergency"
• **Position:** Lithotomy. Buttocks to edge of bed.
• **Team:** EM Reg (lead), Paediatric Reg, Obstetric Reg, Anaesthetics, Neonatal Team

**GOLDEN RULE — HANDS OFF:**
• Do NOT touch baby until umbilicus is visible
• Allow baby to hang — gravity delivers
• Deliver head within **5 min** of buttocks / **3 min** of umbilicus — intervene if delayed

**Paeds Considerations:**
• Adolescent pelvis may be narrower — be prepared for head entrapment
• Concealed pregnancy: gestational age may be unknown — prepare NLS for all ≥22/40
• Warm environment critical — switch on Resuscitaire heater immediately

## EQUIPMENT

- ☐ Chlorhexidine
- ☐ Sterile Drape
- ☐ Neonatal Resuscitation Kit
- ☐ **Resuscitaire (Heater on)** SWITCH ON NOW
- ☐ Oxytocin 10 units IM FETCH DRUG
- ☐ GTN Spray (uterine relaxation) FETCH DRUG
- ☐ Episiotomy Scissors
- ☐ Cord Clamps ×2
- ☐ **Broselow Tape**

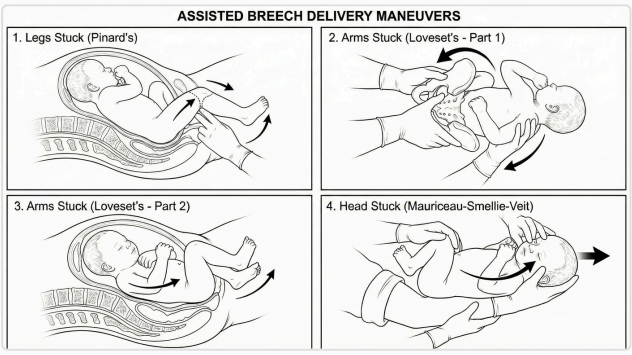
## POST-PROCEDURE

• Inspect perineum for tears (3rd/4th degree)
• Active management 3rd stage: Oxytocin 10 units IM (or 5 units slow IV)
• Paediatric review for baby (hips, limbs, cord)
• Transfer to Obstetric Unit / Theatre for repair

## MANOEUVRES (IF DESCENT STOPS)

1. **Legs Stuck (Pinard's):** Insert finger behind fetal knee (popliteal fossa). Abduct thigh laterally. Knee flexes. Sweep out.
2. **Arms Stuck (Lovset's):** Hold bony pelvis (thumbs on sacrum). Rotate 180° (keep back up). Posterior arm sweeps down. Deliver arm. Rotate back 180° for second arm.
3. **Head Stuck (Mauriceau-Smellie-Veit):** Assistant: Suprapubic pressure. Operator: Baby on forearm. Fingers on malar eminences. Flex head. Pull gently.
4. **Head Trapped — Adolescent Pelvis:** GTN Spray `400mcg` sublingual (1-2 puffs) — uterine relaxation in ~60 sec. Call Obstetric Reg if not present. Consider Dührssen's cervical incisions (specialist only).

**Adolescent Pelvis:** May be android/anthropoid shaped with narrower subpubic angle. Head entrapment is more likely. Have GTN available. Episiotomy alone will NOT resolve head entrapment.




*Pinard's: Abduct thigh to flex knee*

# EMERGENCY BREECH DELIVERY

**Page 2: Action Flowchart — PAEDIATRIC (<16 YRS)**

🩺 PAEDIATRIC EDITION

- **GOLDEN RULE: HANDS OFF | Allow breech to hang | Keep room warm | Head within 5 min of buttocks / 3 min of umbilicus**

## 1 LEGS STUCK (PINARD'S)

- Frank breech. Finger in popliteal fossa. Abduct thigh. Knee flexes. Sweep out.

## 2 ARMS STUCK (LOVSET'S)

- Winged scapula sign. Hold bony pelvis. Rotate 180°. Arm sweeps over chest. Rotate back.

## 3 HEAD STUCK (MAURICEAU-SMELLIE-VEIT)

- Baby on forearm. Fingers on malar. Flex head. Pull gently.

Check cord around neck — slip over head or double-clamp and cut if tight.

## 4 HEAD TRAPPED — ADOLESCENT PELVIS

- Call Obstetric Reg immediately.
- GTN Spray `400mcg sublingual`.
- Consider Dührssen's cervical incisions (specialist only).

🩺 Do NOT apply fundal pressure — worsens impaction.

⚠️ **LAST RESORT**
Zavanelli manoeuvre (cephalic replacement) + Emergency caesarean (Obstetric Reg).

## 5 POST-PROCEDURE

- Inspect tears. Oxytocin `10 units IM` (or 5 units slow IV). FETCH DRUG
- Paediatric review of neonate.

OBSTETRIC EMERGENCY

*Feedback/Updates: jake.turner@uhb.nhs.uk*

# RESUSCITATIVE THORACOTOMY

Page 1: Walkthrough — PAEDIATRIC (<16 YRS)

## INITIAL ACTIONS & INDICATIONS

• Call **2222** "Trauma Cardiac Arrest". Activate Major Haemorrhage Protocol. Request O-neg blood.
• Activate **WMAS/MERIT** immediately — transfer to **Birmingham Children's Hospital (MTC)**.
Contact **KIDS NTS** (advice + retrieval): **0300 200 1100** (24hrs) | backup: **0121 333 9687**

**PAEDIATRIC INDICATIONS (tighter than adult):**
• **PENETRATING** thoracic/epigastric: Arrest <15 mins + signs of life (pupil response, gasping, ECG activity, cardiac motion on US)
• **BLUNT trauma:** Pulseless WITH signs of life + thoracic source on adjuncts (e.g. USS)
• Signs of Life (SOL) = pupils reacting, gasping, ECG activity, cardiac motion on USS

**PAEDIATRIC CONTRAINDICATIONS:**
• Pre-hospital CPR >15 mins after penetrating injury without SOL
• Blunt trauma WITHOUT signs of life (strongly recommended against)
• Severe head injury + cardiac arrest

🩺 PTS/WTA/EAST 2023: EDT conditionally recommended if pulseless WITH signs of life (penetrating, or blunt with thoracic source). Recommended against if NO signs of life.

**Goals:** Release tamponade | Internal cardiac massage | Occlude descending aorta

## EQUIPMENT TRAY

- [ ] Chlorhexidine
- [ ] Sterile Drape / Gown Pack
- [ ] **Scalpel: Size 22**
- [ ] **Tuff Cut Scissors (Heavy Duty)**
- [ ] Gigli Saw (Handles + Wire) or Bone Cutters
- [ ] **Finochietto Retractor**(pliable ribs in young children)
- [ ] Toothed Forceps (Bonney's)
- [ ] Vascular Clamp (Satinsky/DeBakey)
- [ ] Internal Paddles (for Defib)
- [ ] Artery Forceps (Spencer Wells/Kellys)
- [ ] Foley Catheter 12-14F (<30kg), 20F (larger) — last resort
- [ ] Prolene/Silk 2-0 (Horizontal Mattress)
- [ ] Skin Stapler / Large Dressings

## PROCEDURE WALKTHROUGH

1. **Position:** Supine. Left arm extended overhead (exposes left thorax).
2. **Incision:** Left anterolateral thoracotomy — 4th or 5th ICS (below nipple, above 5th rib). Incise from sternum to mid-axillary line.
3. **Enter Chest: Pause ventilation.** Scissors through intercostals (upper border of rib). Finger to guide entry. Avoid lung.
4. **Retract:** Insert Finochietto retractor (or manually retract ribs). In young children, ribs are pliable — may not need saw.
5. **Pericardiotomy:** Grasp pericardium anterior to phrenic nerve. Vertical incision — release tamponade. Remove clot manually.
6. **Internal Cardiac Massage: Flat hands/palm, no fingertips.** Two-handed (or one-handed in small children). 100 compressions/min, allow filling.
7. **Aortic Occlusion:** Descending aorta (left of spine; oesophagus lies next to it — NG tube helps). **Compress against spine first**; clamp (Satinsky) only if trained. Shunts blood to coronary + cerebral.
8. **Defibrillate:** VF/pVT — Internal paddles **10-20J**.

🩺 Paediatric ribs are more pliable — manual spreading may suffice without Gigli saw in children <10 yrs. One-handed cardiac massage for small children.

## POST-PROCEDURE PLAN

If ROSC: Partly close, **retractor in (loosened)** for re-entry. Large dressings. **Keep warm.** Sedate and paralyse. Transfer: **Birmingham Children's Hospital (MTC)** via WMAS/MERIT.
KIDS NTS: **0300 200 1100** for retrieval coordination.

📍 **STORAGE**

**FINOCHIETTO RETRACTOR AND GIGLI SAW IN THORACOTOMY TRAY**

# RESUSCITATIVE THORACOTOMY

**Page 2: Action Flowchart — PAEDIATRIC (<16 YRS)**

- **PENETRATING + SOL = PROCEED | BLUNT + SOL + thoracic source = consider | NO signs of life = do NOT proceed**

🩺 PTS/WTA/EAST 2023: signs of life are the key decision point in children.

## 1 LEFT ANTEROLATERAL THORACOTOMY

- Incision: 4th/5th ICS, sternum to mid-axillary line, below nipple/above 5th rib.
- **Pause ventilation.** Divide intercostals (upper rib border) with Tuff Cut scissors (blade up, finger guide).
- Spread ribs — Finochietto OR manual in young/pliable ribs.

In children <10 yrs, ribs may spread manually without Gigli saw.

## 2 PERICARDIOTOMY & TAMPONADE RELEASE

- Grasp pericardium anterior to phrenic nerve. Vertical incision.
- Open longitudinally. Remove clot manually.
- Internal cardiac massage: 2-handed adults/older teens. 1-handed in small children.

## 3 CARDIAC MASSAGE & DEFIB

- Flat hands/palm, no fingertips. Rate ~100/min, allow filling.
- VF/pVT: Internal paddles — `10-20J (paediatric setting)`.

## 4 AORTIC OCCLUSION

- Identify descending aorta posterior-left of thoracic cavity. **Compress against spine first.**
- Clamp (Satinsky) only if trained. Diverts blood to cerebral/coronary circulation.
- Do NOT release clamp until haemorrhage controlled.

## 5 WOUNDS & CLOSURE

- Cardiac wounds: **finger pressure first**, then suture. Foley (12-14F small children) = last resort — can enlarge defect.
- Lung: counted swabs → non-crushing clamp → hilar clamp if massive haemorrhage.
- ROSC: partly close, retractor in (loosened). Keep warm. Immediate transfer.

STOP CRITERIA: >30 mins CPR without ROSC | No cardiac activity on direct visualisation | Non-survivable injuries



*Feedback/Updates: jake.turner@uhb.nhs.uk*

PTS / WTA / EAST Systematic Review & Practice Management Guideline: ED Thoracotomy in Children | J Trauma Acute Care Surg 2023;95(3):432-441 | Technique: Chesters A et al. Emergency thoracotomy: how to do it in 2026. Emerg Med J 2026

Open Guideline (J Trauma 2023)

# ESCHAROTOMY (BURNS)

Page 1: Walkthrough — PAEDIATRIC (<16 YRS)

## INITIAL ACTIONS & INDICATIONS

• Call **2222** "Burns Emergency". Early contact with Regional Burns Centre.
• **Indication:** Full thickness circumferential burns causing:
– Chest: Respiratory failure (reduced chest expansion, rising pressures)
– Limb: Vascular compromise (absent distal pulse, CRT >2s, pain/pallor/paraesthesia)

**Fluid Resuscitation (start BEFORE escharotomy):**
`PAEDIATRIC Parkland: 3ml × weight(kg) × %TBSA`
(partial + full thickness; exclude erythema) + maintenance fluids
Half in first 8 hours, half over next 16 hours. Use Hartmann's or Plasmalyte.
Urine output target: `>1ml/kg/hr` in children <30kg
Threshold for resus: **>10% TBSA** in children (vs 15% adults)
BBA referral threshold: **≥2% TBSA** in children (vs ≥3% adults)

🩺 TBSA: Use Lund-Browder chart (NOT Rule of Nines — less accurate in children). Child's palm + fingers ≈ 1% TBSA.

## EQUIPMENT

- [ ] Chlorhexidine
- [ ] Sterile Drape / Gown Pack
- [ ] **Scalpel: Size 22 or 10**
- [ ] **Surgical Diathermy** (preferred — reduces bleeding)
- [ ] Large Swabs
- [ ] Dressings (paraffin gauze / Mepitel)
- [ ] Cling film strips — no creams pre-burns centre

## 📍 STORAGE

**ESCHAROTOMY EQUIPMENT ON BURNS TROLLEY**

## POST-PROCEDURE

• Monitor CRT, Doppler pulse, SpO2 every 15 mins
• Re-dress with paraffin gauze / Mepitel
• Transfer to Burns Centre
• Re-evaluate every 2-4 hours — may need repeat escharotomy

## ANATOMICAL LINES — LIMB

**1. Upper Limb:** Medial and lateral incisions along the full length of the limb. Include thenar/hypothenar eminences and dorsal interossei if hand affected.
**2. Lower Limb:** Medial and lateral incisions. Include dorsum of foot if foot affected.

**Critical: Do NOT extend into muscle/fascia (that is fasciotomy — requires surgical referral)**

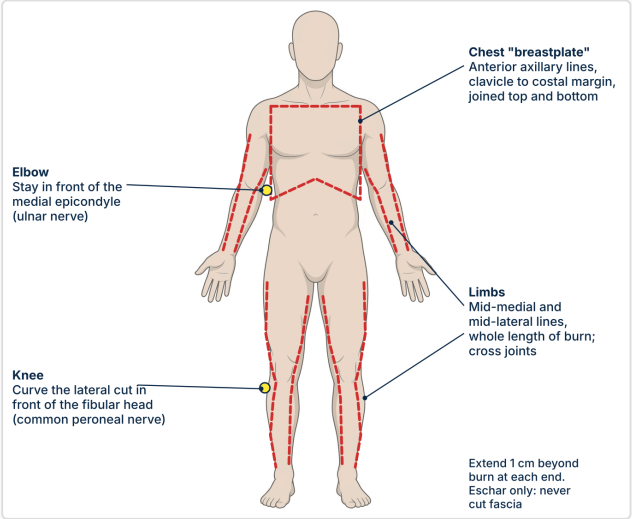
## ANATOMICAL LINES — CHEST

**1.** Bilateral anterior axillary line incisions (vertical, clavicle to costal margin).
**2.** Connect across sternum with transverse subcostal incision if needed.
**3.** May also extend along costal margin bilaterally.

**Incision depth = through full dermis to subcutaneous fat only.**

## PROCEDURE STEPS

1. Analgesia + sedation — children have pain from surrounding partial thickness injury
2. Clean area with chlorhexidine
3. Mark incision lines with pen
4. Scalpel incision through eschar to subcutaneous fat (not muscle)
5. Check for adequate release: limb = palpable pulse returns; chest = improved chest rise
6. Dress with paraffin gauze or Mepitel — do NOT use dry dressings




*Escharotomy lines: medial/lateral limb + anterior axillary chest*

# ESCAROTOMY (BURNS)

Page 2: Action Flowchart — PAEDIATRIC (<16 YRS)

- **INDICATION: Full thickness circumferential burn + respiratory failure OR limb ischaemia | Contact Burns Centre EARLY**

## 1 FLUID RESUSCITATION (FIRST)

- Parkland: `3ml × kg × %TBSA` (Hartmann's/Plasmalyte)
- Half in 8hrs, half over 16hrs (from time of burn)
- Target urine: `>1ml/kg/hr` (<30kg children)

🩺 Threshold ≥10% TBSA for resus in children (adults: 15%)

## 2 CHEST ESCAROTOMY

- Both anterior axillary line incisions — clavicle to costal margin
- Transverse subcostal connection if needed
- Depth: Dermis to fat only — NOT muscle
- Check: Improved chest rise and ventilator pressures

## 3 LIMB ESCAROTOMY — MEDIAL/LATERAL

- Medial AND lateral incisions — full length of affected limb
- Include hand/foot if involved
- Depth: Dermis to fat — NOT into fascia
- Check: Return of palpable pulse + Doppler signal

## 4 CONFIRM RELEASE & DRESS

- Limb: Palpable pulse, CRT <2s, Doppler signal
- Chest: Improved chest expansion, SpO2, reduced ventilator pressures
- Dress with paraffin gauze / Mepitel

⚠️ **IF RELEASE INADEQUATE**

Fasciotomy required (surgical specialty). Do NOT attempt blind fasciotomy in ED.

Burns Centre contact mandatory. BBA referral: ≥2% TBSA in children. Do NOT delay transfer.



*Feedback/Updates: jake.turner@uhb.nhs.uk*

# CHEST DRAIN (OPEN)

Page 1: Walkthrough — PAEDIATRIC (<16 YRS)

## INITIAL ACTIONS & INDICATIONS

**INDICATIONS:** Tension pneumothorax (needle decompression first), Haemothorax, Massive haemothorax, Traumatic pneumothorax

- Needle decompression FIRST if tension: 2nd ICS mid-clavicular line OR 4th/5th ICS anterior axillary line
- **Triangle of Safety:** anterior border of latissimus dorsi, lateral border of pectoralis major, superior to nipple, below axilla
- Insert ABOVE the rib (avoid neurovascular bundle on inferior rib border)
- Ultrasound guidance strongly recommended where time allows

**PAEDIATRIC SIZE GUIDE:**

`Chest drain (French) = 4 × uncuffed ETT size`

Newborn: 8–12 Fr | Infant (<1yr): 12–16 Fr
Child (1–12 yrs): 12–24 Fr | Adolescent (>12 yrs): 12–32 Fr

**NEVER use trochar technique in children**

## EQUIPMENT

- ☐ Chlorhexidine
- ☐ Sterile Drape / Gown Pack
- ☐ Scalpel: Size 10 or 15
- ☐ Blunt dissection forceps (artery clip)
- ☐ **Chest drain — appropriately sized**(see size guide)
- ☐ Underwater seal drainage system
- ☐ Large bore suction
- ☐ Suture (1-0 Prolene / Silk)
- ☐ Occlusive dressing
- ☐ Lignocaine 1% (max `3mg/kg`)

## 📍 STORAGE

**CHEST DRAIN EQUIPMENT ON TRAUMA TROLLEY**

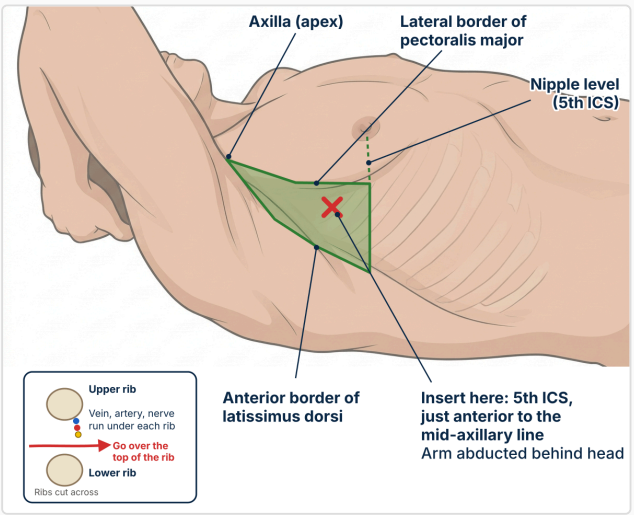
## POST-PROCEDURE

- CXR to confirm position
- Massive haemothorax: >15–20 ml/kg initially or 3–4 ml/kg/hr → MHP + urgent surgical review
- Secure drain — never to clothing
- Suction: -5 to -10 cmH2O

## PROCEDURE WALKTHROUGH

1. **Position:** Arm on affected side abducted and elevated above head.
2. **Identify:** 5th ICS, just anterior to the mid-axillary line (Triangle of Safety). Mark with pen.
3. **Clean:** Chlorhexidine. Drape. Infiltrate lignocaine — skin, rib, pleura (max `3mg/kg` in children).
4. **Skin Incision:** Horizontal incision parallel to rib, 1.5-2cm. Into subcutaneous fat.
5. **Blunt Dissection:** Artery clip — dissect over superior border of rib into pleural space. Feel 'give' on entry.
6. **Finger Sweep:** In older children (>8 yrs) — single finger sweep to confirm pleural space. In small children — NOT possible; use forceps to guide drain.
7. **Insert Drain:** Feed drain WITHOUT trochar. Aim apex for air, base for fluid.
8. **Connect:** Underwater seal. Confirm: Swinging + fogging of tube.
9. **Secure:** Suture. Occlusive dressing. Mark drain at skin level.

🩺 No finger sweep in small children — insert with forceps guidance only. No trochar technique in any paediatric patient.




*Triangle of Safety — 5th ICS, just anterior to the mid-axillary line. Finger sweep only in children >8 yrs*

# CHEST DRAIN (OPEN)

**Page 2: Action Flowchart — PAEDIATRIC (<16 YRS)**

- **SIZE: 4×ETT size (French) | Triangle of Safety: 5th ICS, just anterior to the mid-axillary line | NEVER use trochar in children**

## 1 NEEDLE DECOMPRESSION (IF TENSION SUSPECTED)

- 2nd ICS mid-clavicular line OR 4th/5th ICS anterior axillary line
- Use 14-16G cannula (standard length 3.8cm sufficient for all paeds — thinner chest wall)
- Remove needle, leave cannula. Definitive drain follows.

> 🩺 In small children, needle decompression may need repeating — chest wall thinner and cannula kinks easily.

## 2 SKIN INCISION

- Pause 5 seconds. Confirm triangle of safety landmarks.
- Horizontal incision, 1.5-2cm, parallel to rib, above lower rib.
- LA max: `3mg/kg lignocaine` (NOT 5mg/kg — children more sensitive)

## 3 BLUNT DISSECTION & ENTRY

- Artery clip over superior rib border into pleural space.
- Feel 'give'. Do NOT use trochar.
- Older children (>8): single finger sweep. Young children: forceps only.

## 4 INSERT DRAIN & CONNECT

- Feed drain WITHOUT trochar. Aim: apex = air, base = fluid.
- Connect underwater seal. Confirm swing/fogging.
- Suture and dress. Mark at skin level.

> >15–20 ml/kg initially or 3–4 ml/kg/hr → urgent surgical/thoracic review.

## 5 CONFIRM & ESCALATE

- CXR to confirm position | Suction: -5 to -10 cmH2O | Monitor hourly drainage output

Massive haemothorax (>15–20 ml/kg initially or 3–4 ml/kg/hr ongoing): Activate Major Haemorrhage Protocol. Urgent surgical/thoracic review — consider thoracotomy.



*Feedback/Updates: jake.turner@uhb.nhs.uk*

# PERICARDIOCENTESIS

Page 1: Walkthrough — PAEDIATRIC (<16 YRS)

## INITIAL ACTIONS & INDICATIONS

**INDICATION:** CLINICAL tamponade (dyspnoea, raised JVP, hypotension, tachycardia, pulsus paradoxus) with haemodynamic compromise or arrest — **drain urgently**. Echo signs alone are NOT an indication.
**Contraindications:** none absolute in shock. Aortic dissection / free-wall rupture → surgery. Relative: coagulopathy, anticoagulants, platelets <50.
**Trauma:** failed aspiration does NOT exclude haemopericardium (clot).
• **ULTRASOUND-GUIDED** preferred at all times — blind technique only in arrest/immediately life-threatening
• USS findings: Pericardial effusion with right ventricular diastolic collapse
• In children: USS subxiphoid or parasternal view. Measure skin-to-pericardium depth on USS.

## EQUIPMENT

- [ ] Chlorhexidine
- [ ] Sterile Drape / Gown Pack
- [ ] **18G spinal needle**or Seldinger pericardiocentesis kit
- [ ] 10/20ml Syringe with 3-way tap
- [ ] Ultrasound machine (if available)
- [ ] ECG monitoring (continuous)
- [ ] Guidewire (if Seldinger technique)
- [ ] Agitated saline (5ml) — confirm position
- [ ] Pigtail catheter (if indwelling drain)
- [ ] Lignocaine 1% (max `3mg/kg`)

## 📍 STORAGE

**PERICARDIOCENTESIS KIT — CHECK LOCAL STORAGE LOCATION**

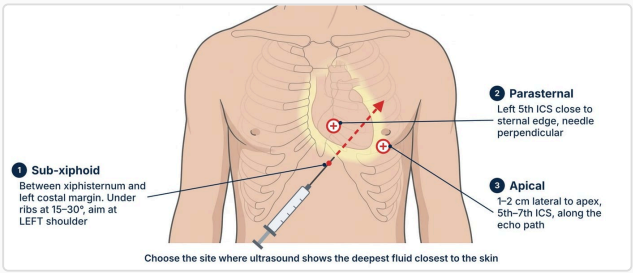
## POST-PROCEDURE

• CXR (exclude pneumothorax). Reassess ECG + USS
• Pigtail: aspirate every 4–6h; remove when <25–30ml/24h
• ICU admission
• Monitor for recurrence (especially in malignancy, sepsis)

## PROCEDURE WALKTHROUGH

1. **Position:** Semi-reclining ~30°, slightly rotated to LEFT.
2. **USS:** Entry site where the pericardial space is closest to the probe and fluid is deepest, with no organs in the way. Mark on skin.
3. **Clean:** Chlorhexidine. Sterile drape. LA at puncture site (`3mg/kg max`).
4. **Needle Insertion (Subxiphoid):** 18G needle, subxiphoid angle — between xiphoid and left costal margin. Under ribs, lower to 15–30°, aim LEFT shoulder (too steep → peritoneum; too medial → RA). Advance with continuous aspiration. USS-guided: watch needle tip enter pericardium.
5. **Confirm & drain:** 5ml agitated saline — bubbles in pericardium. Drain gradually; stop once haemodynamics normalise.
6. **Seldinger:** If indwelling drain needed — introduce guidewire, remove needle, dilate, insert pigtail catheter.
7. **Secure:** Suture. Connect to 3-way tap.

🩺 Skin-to-pericardium depth varies with age — measure on USS and do not advance beyond it. Use 18G needle (NOT larger).




*Anatomy & approaches — this page uses subxiphoid: 15–30° towards left shoulder*

# PERICARDIOCENTESIS

**Page 2: Action Flowchart — PAEDIATRIC (<16 YRS)**

- **CLINICAL tamponade + compromise = DRAIN (echo signs alone are not an indication) | Hypotension, tachycardia, dyspnoea, raised JVP, pulsus paradoxus | USS: RA/RV diastolic collapse**

## 1 USS-GUIDED ASSESSMENT

- Confirm effusion on USS. RV diastolic collapse = tamponade.
- Entry: closest pericardial space + deepest fluid, no organs in the way. Mark on skin.
- Temporise: fluid `10ml/kg`, NO diuretics, dobutamine if inotrope needed, avoid positive-pressure ventilation unless essential.

In cardiac arrest — proceed without USS. Subxiphoid: 15–30° towards left shoulder.

## 2 NEEDLE INSERTION (SUBXIPHOID)

- Pause 5 seconds. Confirm landmarks.
- 18G needle at xiphoid-costal margin junction.
- Under ribs, lower to 15–30°, aim LEFT shoulder.
- Advance with continuous aspiration. USS-guided: watch needle tip. Stop on aspiration of fluid.

## 3 ASPIRATION

- Confirm with 5ml agitated saline — bubbles in pericardial space. Bloody tap = myocardial puncture OR haemorrhagic effusion.
- Drain gradually; stop once haemodynamics normalise.
- 3-way tap — alternate aspiration and closure.

## 4 SELDINGER (IF INDWELLING DRAIN NEEDED)

- Introduce guidewire through needle. Remove needle.
- Dilate tract. Insert pigtail catheter.
- Suture and connect to closed drainage system.

🩺 Depth varies with age — measure on USS. 18G needle only.

FAILURE/DETERIORATION: Consider cardiac injury → emergent thoracotomy. Failed aspiration does NOT exclude haemopericardium (clot).



*Feedback/Updates: jake.turner@uhb.nhs.uk*

# EXTERNAL (TRANSCUTANEOUS) PACING

Page 1: Walkthrough — PAEDIATRIC (<16 YRS)

## INITIAL ACTIONS & INDICATIONS

**INDICATIONS:** Symptomatic bradycardia not responding to atropine/adrenaline | Complete heart block | Drug-induced bradycardia (beta-blocker, digoxin, CCB overdose)

**CONTRAINDICATIONS:** Hypothermia as primary cause (rewarm first) | Asystole in non-shockable cardiac arrest

**PAEDS PACING SETTINGS:**
Rate: `80-100 bpm` (NOT adult default 60)
mA: Start at `10mA`. Increase in 10mA increments until electrical capture
Output: Set `5-10mA ABOVE` capture threshold
Mode: DEMAND if intrinsic rhythm present. NON-DEMAND during CPR or no intrinsic rhythm

**PAD SIZE:**
<10kg: **Paediatric combo pads**
>10kg: Adult combo pads acceptable
**Placement:** Anterior-posterior preferred (right of sternum + left posterior subscapular)

## EQUIPMENT

- [ ] Defibrillator/pacemaker unit
- [ ] **Paediatric combo pads**(if <10kg)
- [ ] Limb leads (4-lead ECG monitoring)
- [ ] Backup IV/IO access
- [ ] **Analgesia + sedation**(pacing is painful)

## POST-PROCEDURE

• Confirm mechanical capture: palpate femoral pulse for each paced beat
• Monitor BP
• Arrange urgent cardiology review
• Transfer to PICU/Cardiology

## PROCEDURE STEPS

1. **MONITORING:** Attach 4-lead ECG. Confirm bradycardia. Identify intrinsic rate and P-wave activity.
2. **ANALGESIA:** If conscious — IV/IO analgesia NOW before pacing. Morphine `0.1mg/kg IV` or Ketamine `0.5-1mg/kg IV`
3. **PADS:** Apply combo pads. <10kg = paediatric pads. >10kg = adult pads. Anterior-posterior preferred.
4. **SET RATE:** `80-100 bpm`
5. **SET mA TO ZERO:** Confirm pacemaker recognising QRS complexes.
6. **INCREASE mA:** Start at 10mA. Increase by 10mA increments. Electrical capture = pacing spike + wide QRS on ECG.
7. **SET FINAL OUTPUT:** Set `5-10mA ABOVE` capture threshold.
8. **CONFIRM MECHANICAL CAPTURE:** Palpate pulse for each captured beat. DO NOT rely on ECG alone.

🩺 Pacing is painful. Do not delay analgesia in a conscious child. External pacing is a bridge — arrange definitive rhythm management urgently.

🩺 In CPR — use NON-DEMAND mode (chest compressions interfere with QRS sensing in demand mode).

# EXTERNAL (TRANSCUTANEOUS) PACING

Page 2: Action Flowchart — PAEDIATRIC (<16 YRS)

- **PACING RATE: 80-100 bpm | START at 10mA, increase in 10mA steps | PADS: <10kg = paediatric pads | CONFIRM MECHANICAL capture — not just electrical**

## 1 PREPARE & MEDICATE

- 4-lead ECG monitoring. Confirm bradycardia.
- Analgesia if conscious: Morphine `0.1mg/kg IV` OR Ketamine `0.5-1mg/kg IV` GIVE NOW
- Apply combo pads: <10kg = paediatric, >10kg = adult. Anterior-posterior preferred.

## 2 SET UP PACER

- Set mode: DEMAND if intrinsic rhythm. NON-DEMAND during CPR.
- Set rate: `80-100 bpm`.
- Set mA to ZERO initially. Confirm pacemaker recognises QRS.

## 3 ACHIEVE CAPTURE

- Increase mA from 10mA in 10mA increments.
- ELECTRICAL capture = pacing spike + wide QRS complex on ECG.
- Set final output `5-10mA ABOVE` threshold.

In young children, max output may be needed — do not stop early.

## 4 CONFIRM MECHANICAL CAPTURE

- Palpate femoral pulse — confirm pulse for each paced beat.
- Do NOT rely on ECG trace alone.
- If no mechanical capture despite electrical capture → check output, reposition pads.

## 5 ESCALATE

- Arrange urgent cardiology review. Transfer to PICU. Monitor BP.

🩺 External pacing is a bridge only. Arrange transvenous or internal pacing urgently.

⚠️ **NO CAPTURE AT MAXIMUM MA**

Reposition pads to anterior-posterior. Ensure good skin contact. Adrenaline `1mcg/kg` IV (emergency only). If no pulse → cardiac arrest algorithm.

Hypothermia primary cause → STOP pacing. Prioritise rewarming. Pacing will not work in severe hypothermia.



*Feedback/Updates: jake.turner@uhb.nhs.uk*

# SUBCLAVIAN / BRACHIOCEPHALIC CENTRAL LINE

Page 1: Walkthrough — PAEDIATRIC (<16 YRS)

## INITIAL ACTIONS & INDICATIONS

**INDICATION IN PAEDS:** Massive haemorrhage requiring rapid large-bore central access. NOT for cardiac arrest (use IO access in cardiac arrest).

**PAEDIATRIC PREFERRED APPROACH:**
- USS-guided **supraclavicular brachiocephalic vein** (jugulo-subclavian junction) — recommended by European societies for children
- Internal jugular (USS-guided): High risk of back-wall puncture/transfixion in infants and small children — use with caution
- Infraclavicular subclavian: Higher complication rate in small children; acceptable in older adolescents
- **USS MANDATORY** in paediatric patients

**SIZE GUIDE:**
<10kg: 4Fr | 10–20kg: 4.5–5Fr | 20–40kg: 5Fr (double lumen) | >40kg: 7Fr

**INSERTION DEPTH (Andropoulos 2001):**
Height ≤100cm: (height/10) – 1 cm | >100cm: (height/10) – 2 cm. Confirm tip on CXR.

## EQUIPMENT

- [ ] Chlorhexidine + Sterile Drape
- [ ] **Ultrasound machine + sterile cover**
- [ ] Central line kit: appropriate French size
- [ ] LA: Lignocaine 1% (max `3mg/kg`)
- [ ] 10ml syringe with normal saline flush
- [ ] Suture (2-0 silk or Prolene)
- [ ] CVC dressing kit

## 📍 STORAGE

**CENTRAL LINE KIT — CHECK LOCAL STORAGE LOCATION**

## POST-PROCEDURE

- CXR: exclude pneumothorax; tip lower SVC / cavo-atrial junction
- **"Wire out, shout out"** — witness confirms, document
- Document insertion depth (cm at skin)
- Convert to IO access for cardiac arrest — central line placement wastes time

## USS-GUIDED SUPRACLAVICULAR APPROACH

1. **Position:** 10-15° Trendelenburg. Head turned slightly away from puncture site.
2. **USS:** Identify brachiocephalic vein (jugulo-subclavian junction) in supraclavicular fossa. Identify subclavian artery (pulsatile).
3. **Clean:** Chlorhexidine. Sterile drape. Sterile USS cover.
4. **LA:** Infiltrate subcutaneous tissue. Max `3mg/kg` lignocaine.
5. **Needle:** Insert in-plane with USS beam. Track needle tip into vein. Confirm venous position (non-pulsatile, compressible on USS).
6. **Aspirate:** Free blood return in syringe.
7. **Guidewire:** Insert J-wire through needle. Remove needle. **Before dilating:** scan along wire in long axis — confirm it stays in the vein and does not transfix another vessel.
8. **Dilate:** Skin nick with scalpel. Insert dilator over wire (do NOT advance more than needed).
9. **Line:** Railroad catheter over wire to appropriate depth (see depth guide). Remove wire.
10. **Aspirate + flush** all lumens. Suture and dress. CXR.

🩺 USS is mandatory. Brachiocephalic (supraclavicular) approach preferred over IJ in children <5yrs due to high IJ transfixion risk. Subclavian landmark technique not recommended in children.

# SUBCLAVIAN / BRACHIOCEPHALIC CENTRAL LINE

**Page 2: Action Flowchart — PAEDIATRIC (<16 YRS)**

- **IO ACCESS for cardiac arrest — do NOT delay with CVC insertion | USS MANDATORY in paediatric central line insertion | Supraclavicular brachiocephalic approach preferred in children**

## 1 POSITIONING & USS ASSESSMENT

- Trendelenburg 10-15°. Head turned away from side.
- USS: Identify brachiocephalic vein + subclavian artery (avoid). Confirm vein compressibility.
- Size: Use appropriate French size for age (see walkthrough).

## 2 NEEDLE INSERTION (USS IN-PLANE)

- Pause 5 seconds. Confirm landmarks on USS.
- Insert needle in-plane. Track tip to vein on USS.
- Aspirate freely — confirm non-pulsatile blood return.

🩺 Arterial puncture with needle — WITHDRAW, firm pressure ≥5 mins. Do NOT dilate. If dilator/catheter (≥6Fr) already in artery — **LEAVE IN**, urgent paediatric surgeon / IR.

## 3 SELDINGER TECHNIQUE

- Insert J-wire through needle. Remove needle.
- Scan along wire (long axis) — confirm it stays in the vein **before dilating**. Skin nick. Dilate over wire carefully.
- Railroad catheter to appropriate depth (see size guide).

## 4 CONFIRM & SECURE

- Aspirate + flush all lumens. **"Wire out, shout out"** (witnessed). Suture and dress.
- CXR mandatory — exclude pneumothorax; tip lower SVC / cavo-atrial junction.

Mark insertion depth on catheter at skin level with steristrip. Document in notes.

Complications: Pneumothorax (CXR mandatory) | Arterial puncture (withdraw, pressure) | Line malposition (confirm on CXR before use) | Air embolism (Trendelenburg position during insertion reduces risk)



*Feedback/Updates: jake.turner@uhb.nhs.uk*

# LATERAL CANTHOTOMY & CANTHOLYSIS

Page 1: Walkthrough — PAEDIATRIC (<16 YRS)

## INITIAL ACTIONS & INDICATIONS

**INDICATION:** Orbital Compartment Syndrome (OCS) — retrobulbar haemorrhage causing:
- IOP >40 mmHg (normal <20 mmHg)
- Proptosis
- Relative afferent pupillary defect (RAPD)
- Reduced visual acuity
- Taut eyelids / chemosis

**MECHANISM:** Blunt orbital trauma (most common). Also: infection, burns, post-procedural.

**TIME-CRITICAL:** Permanent vision loss after 90–120 minutes of retinal ischaemia

🩺 Technique identical to adult. Procedural sedation MORE IMPORTANT in children — a moving patient risks globe injury.

**ABSOLUTE CONTRAINDICATION:** Suspected globe rupture (irregular pupil, hyphema, laceration, low IOP <5mmHg)

## EQUIPMENT

- ☐ Chlorhexidine
- ☐ Topical anaesthetic eye drops (tetracaine/proxymetacaine)
- ☐ LA: Lignocaine 1% with adrenaline — 2ml at lateral canthus
- ☐ **Straight haemostat (artery clip)**
- ☐ **Small sterile scissors (iris scissors preferred)**
- ☐ Toothed forceps
- ☐ Tonometer (Tono-Pen) if available

## 📍 STORAGE

**CANTHOTOMY EQUIPMENT — CHECK LOCAL STORAGE LOCATION**

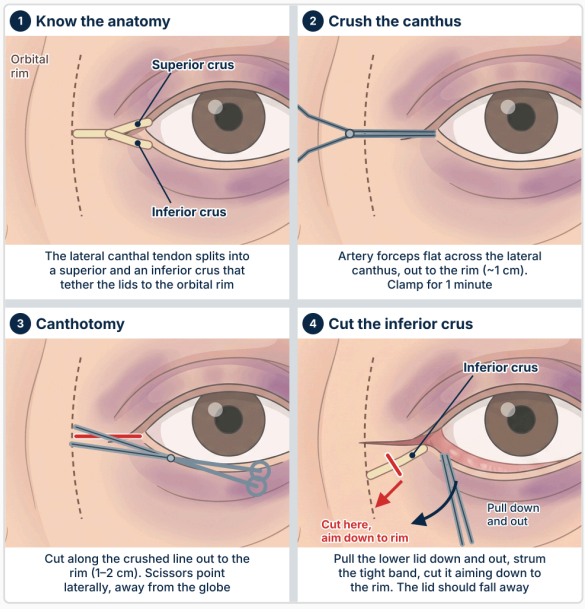
## POST-PROCEDURE

- Re-measure IOP — should fall significantly
- If IOP still elevated: proceed to superior cantholysis
- Ophthalmology referral (urgent — same day or within hours)
- Wound heals spontaneously — no formal repair needed acutely

## PROCEDURE STEPS

1. **ANALGESIA/SEDATION:** Procedural sedation if child is not cooperative (vital — avoid globe injury). Ketamine `1-2mg/kg IV`.
2. **TOPICAL ANAESTHESIA:** Apply tetracaine drops to both eyes.
3. **LOCAL ANAESTHESIA:** 2ml lignocaine 1% with adrenaline at lateral canthus. Inject parallel to globe surface — keep needle away from orbit.
4. **HAEMOSTAT CRUSH:** Apply straight haemostat horizontally at lateral canthus for 30-60 seconds — devascularises tissue, reduces bleeding.
5. **CANTHOTOMY:** Remove haemostat. Cut along crushed tissue from lateral canthus to lateral orbital rim (~1cm). Scissors aimed AWAY from globe (towards lateral orbital rim).
6. **IDENTIFY TENDON:** Pull lower eyelid down and out. Palpate/strum inferior crus of lateral canthal tendon with scissors tip.
7. **CANTHOLYSIS:** Cut inferior canthal tendon — scissors pointed DOWN and AWAY from orbit. Lower lid should fall away.
8. **CHECK:** Lower lid falls away loosely; IOP falls and proptosis/tension reduce.

🩺 Child cooperation is critical — do not perform this without sedation if child is moving. Ketamine procedural sedation is first-line.




*Canthotomy then cantholysis: inferior crus first*

# LATERAL CANTHOTOMY & CANTHOLYSIS

Page 2: Action Flowchart — PAEDIATRIC (<16 YRS)

- **IOP >40mmHg + proptosis/RAPD/vision loss = PROCEED | Globe rupture = CONTRAINDICATED | WINDOW: 90–120 mins before permanent vision loss | SEDATE CHILD before procedure**

## 1 CONFIRM INDICATION

- IOP >40mmHg with: proptosis OR RAPD OR vision loss
- Exclude globe rupture (irregular pupil, hyphema, laceration, IOP <5mmHg)
- Procedural sedation: Ketamine `1-2mg/kg IV` GIVE NOW
- Topical anaesthetic drops + 2ml lignocaine 1%+adrenaline lateral canthus.

## 2 HAEMOSTAT CRUSH

- Apply straight haemostat horizontally — lateral canthus to lateral orbital rim.
- Crush for 30-60 seconds. Devascularises tissue.

🩺 Ensure scissors ALWAYS point away from globe and towards lateral orbital rim.

## 3 CANTHOTOMY (1CM INCISION)

- Remove haemostat.
- Cut along crushed tissue: lateral canthus to lateral orbital rim (~1cm).
- Scissors AWAY from globe — towards lateral rim.

## 4 CANTHOLYSIS (CUT INFERIOR TENDON)

- Pull lower lid down and out. Strum inferior crus of lateral canthal tendon.
- Scissors pointed DOWN and AWAY. Sever inferior canthal tendon.
- Lower lid falls away = successful cantholysis.

## 5 CHECK IOP & ESCALATE

- Re-measure IOP — should fall significantly; orbit/globe feels less tense.
- If IOP still high → Superior cantholysis (caution: lacrimal artery risk).

Wound heals spontaneously. No immediate repair needed. Refer ophthalmology same day.

⚠️ **IOP REMAINS ELEVATED AFTER INFERIOR CANTHOLYSIS**

Superior cantholysis: cut superior crus (increased bleeding risk). If still fails → Surgical orbital decompression (ophthalmology/maxfax).

OPHTHALMOLOGY EMERGENCY

# SEVERE MAX-FAX BLEED

Page 1: Walkthrough — PAEDIATRIC (<16 YRS) | AIRWAY IS THE PRIORITY

## INITIAL ACTIONS & AIRWAY MANAGEMENT

**PAEDIATRIC PRINCIPLE:** Airway management is MORE difficult in paediatric maxillofacial trauma — smaller anatomy, less space, active bleeding

**Causes:** Facial fractures, post-tonsillectomy bleed (common in paeds), dental trauma, penetrating injury, epistaxis (posterior)

- Call **2222** "Maxillofacial Emergency" + ENT/OMFS
- Call: Anaesthetics (difficult airway must be anticipated)
- **Position:** Upright and leaning forward if conscious — allows blood to drain forward
- Do NOT put child supine if they are protecting airway by sitting forward
- **C-spine:** spinal motion restriction as per trauma assessment, but do NOT prioritise it over the airway — no collar in a conscious child who needs to sit forward

**AIRWAY MANAGEMENT:**
- If patient maintaining airway — allow them to sit forward and spit blood — do NOT force supine
- If compromised: RSI with video laryngoscopy (paediatric blade) — anticipate difficult grade
- ETT: cuffed, sized `(Age÷4)+3.5`
- Have surgical airway equipment (eFONA/tracheotomy kit) immediately available

## EQUIPMENT

- [ ] Yankauer suction ×2
- [ ] Wide-bore suction
- [ ] Nasopharyngeal airway — NOT if midface/base-of-skull fracture suspected
- [ ] Video laryngoscopy (paediatric blade)
- [ ] Cuffed ETT — age-appropriate
- [ ] Rapid Rhino 7.5cm bilateral nasal packs
- [ ] Foley catheter 8-10F (posterior nasal tamponade)
- [ ] Bite blocks ×2 (sized to child's mouth)
- [ ] Semi-rigid cervical collar — correctly sized
- [ ] Xylometazoline 0.05% (<12yrs) or 0.1% (>12yrs)
- [ ] TXA: `15mg/kg IV (max 1g)`
- [ ] Silver nitrate sticks

## 📍 STORAGE

**MAXFAX EMERGENCY TROLLEY IN RESUS**

## ANATOMICAL CONTROL POINTS

**1. ANTERIOR EPISTAXIS (Little's Area):**
Direct pressure + silver nitrate cautery. Anterior nasal pack (BIPP/Rapid Rhino).

**2. POSTERIOR EPISTAXIS:**
Rapid Rhino bilateral packs (7.5cm). If not available: Foley 8-10F via nostril: inflate 10ml water, pull forward. Pack anterior with BIPP or Rapid Rhino.

**3. FRACTURED ZYGOMA / MAXILLA:**
Direct pressure to soft tissues. Mobile maxilla → **midface splint (below)**. Manual reduction of Le Fort fractures can reduce arterial bleeding (senior maxfax only). Angiographic embolisation if available (tertiary centre).

**4. OROPHARYNGEAL / TONSILLAR BLEED:**
Direct suction + pack with damp gauze. Adrenaline-soaked gauze (1:10,000) for local vasospasm. Post-tonsillectomy: commonest maxfax bleed in paediatric ED — reactionary (within 6hrs) usually venous, responds to pressure; secondary (day 5-10) often more significant; arterial = immediate theatre.

**Drug Doses:**
TXA: `15mg/kg IV over 10 mins (max 1g)` — give early
Xylometazoline: 0.05% spray (<12yrs), 0.1% (>12yrs)
Adrenaline soaked gauze: 1:10,000 topical

## MIDFACE SPLINT + C-SPINE COLLAR

**WHEN:** Ongoing bleeding from unstable midface (Le Fort) fracture — **only after intubation. HOW, in order:**
1. Nasal balloons (Epistat/Rapid Rhino/Foley) both sides along floor of nose — **do NOT inflate yet**
2. Bite blocks between molars both sides — ETT in the groove, not compressed
3. Apply semi-rigid collar (sized) — splints the mandible
4. Inflate posterior balloons **both sides at once** + gentle traction
5. Inflate anterior balloons **both sides at once**

**Why:** collar holds mandible → blocks hold palate against it → balloons fix maxilla. One-sided inflation can displace fractures.

⚠️ Adult volumes 10ml post / 30ml ant — use device-specific volumes in children. Don't over-tighten collar.

## POST-PROCEDURE

- Definitive control: Operating theatre with ENT/OMFS — this is a bridge
- Imaging: CT face if facial fractures suspected
- Major Haemorrhage Protocol if significant blood loss
- Transfer to ENT/OMFS or Paediatric Surgical unit

# SEVERE MAX-FAX BLEED

**Page 2: Action Flowchart — PAEDIATRIC (<16 YRS)**

- **AIRWAY FIRST — anticipate difficult intubation in all paediatric maxfax bleeding | Sit forward if conscious — do NOT force supine | Have surgical airway ready**

## 1 SECURE AIRWAY

- Conscious + protecting airway: Sit forward. Suction. High-flow O2.
- Compromised: RSI with video laryngoscopy (paediatric blade). ETT = `(Age÷4)+3.5`
- Have eFONA/tracheotomy kit open and ready.

🩺 If child is sitting forward draining blood — this is working. Do not force supine.

## 2 HAEMORRHAGE CONTROL — ANTERIOR EPISTAXIS

- Direct pressure: pinch soft part of nose for 10-15 mins
- Silver nitrate to Kiesselbach's plexus if visible
- Anterior pack: Rapid Rhino or BIPP (lubricated, along floor of nose)

## 3 HAEMORRHAGE CONTROL — POSTERIOR EPISTAXIS

- Rapid Rhino bilateral 7.5cm packs
- If not available: Foley 8-10F via nostril. Inflate 10ml water. Pull forward. Pack anterior space.
- TXA: `15mg/kg IV over 10 mins (max 1g)` GIVE NOW

## 4 UNSTABLE MIDFACE (LE FORT) — PACK & SPLINT (INTUBATED ONLY)

- ① Nasal balloons both sides — do NOT inflate → ② Bite blocks between molars both sides (ETT in groove) → ③ Semi-rigid C-spine collar to splint mandible
- ④ Inflate posterior balloons both sides simultaneously + gentle traction → ⑤ Inflate anterior balloons both sides simultaneously
- Adult volumes up to 10ml posterior / 30ml anterior — use device-specific volumes in children. Check collar fit.

🩺 Conscious child sitting forward: no collar, no supine — spinal motion restriction never takes priority over the airway.

## 5 TONSILLAR / OROPHARYNGEAL BLEED

- Suction + direct pressure with damp gauze
- Adrenaline 1:10,000 soaked gauze — topical vasospasm
- Post-tonsillectomy: reactionary (within 6hrs) usually responds to pressure; secondary (day 5-10) = likely theatre

## 6 ESCALATE

- Major Haemorrhage Protocol if significant loss. Urgent ENT/OMFS review.
- CT face if fractures suspected. Transfer to theatre or tertiary centre.

⚠️ **UNCONTROLLED POSTERIOR EPISTAXIS DESPITE PACKING**

Angiographic embolisation or surgical ligation. Transfer to ENT/OMFS tertiary centre urgently.

SPECIALIST EMERGENCY

*Feedback/Updates: jake.turner@uhb.nhs.uk*

RCEM Learning Max-Fax (Floating in the Face of Danger) | GNAAS Maxillofacial Trauma SOP v4 (2022) | Harris et al. Eur J Emerg Med 2010;17:230-3 | BAOMS

# LAST — LOCAL ANAESTHETIC SYSTEMIC TOXICITY

**Page 1: Walkthrough — PAEDIATRIC (<16 YRS)**

## INITIAL ACTIONS & RECOGNITION

**RECOGNITION:** Symptoms after local anaesthetic administration

**CNS Features (early):** Metallic taste, circumoral numbness, tinnitus, confusion, agitation
**CNS Features (late):** Seizures, coma, respiratory arrest
**CVS Features:** Widened QRS, QTc prolongation, ventricular arrhythmias, bradycardia, cardiac arrest

**Highest risk drugs:** Bupivacaine >> Levobupivacaine > Ropivacaine > Lignocaine

**MAXIMUM SAFE DOSES (per ideal body weight):**
Bupivacaine: `2mg/kg` (with or without adrenaline)
Lignocaine: `3mg/kg` (alone) | `5mg/kg` (with adrenaline)
Ropivacaine: `3mg/kg`
Prilocaine: `5mg/kg` (alone) | `7mg/kg` (with adrenaline)

Toxicity can occur BELOW these doses with intravascular injection

## EQUIPMENT & DRUGS

- [ ] **Intralipid 20%**— from LAST box/trolley
- [ ] IV access (or IO)
- [ ] Standard resuscitation equipment

**Propofol is NOT a substitute for lipid | In arrest: adrenaline ≤1mcg/kg, avoid vasopressin**
**Consider: Sodium bicarbonate 1-2 mmol/kg for wide QRS/refractory arrhythmias (not in AAGBI QRH)**

## 📍 STORAGE

**INTRALIPID 20% — LAST BOX LOCATION: [FILL IN LOCAL LOCATION]**

## POST-PROCEDURE

• ICU admission — lipid effect wears off (may re-arrest)
• Monitor: ECG, cardiac enzymes. Exclude pancreatitis — daily amylase/lipase. Report as critical incident.
• Contact anaesthetics + PICU

## PAEDIATRIC INTRALIPID DOSES

**Bolus:** `1.5ml/kg IV over 2–3 min` (repeat at 5 and 10 min if not stable; max 3 boluses)
**Infusion:** `15ml/kg/hour` → double to `30ml/kg/hour` if unstable
**Maximum cumulative dose:** `12ml/kg`

**Example weights:**
10kg: Bolus 15ml | Infusion 150ml/hr | Max 120ml total
20kg: Bolus 30ml | Infusion 300ml/hr | Max 240ml total
30kg: Bolus 45ml | Infusion 450ml/hr | Max 360ml total

## PROCEDURE WALKTHROUGH

1. STOP local anaesthetic immediately (remember infusion pumps)
2. CALL: 2222. Activate full resuscitation team. Anaesthetics + PICU urgently.
3. PALS as appropriate — continue resuscitation throughout
4. AIRWAY: 100% O2. Intubate if needed. Avoid hypercarbia — consider mild hyperventilation.
5. SEIZURES: Small incremental benzodiazepine (drug of choice) — midazolam `0.1mg/kg IV/IO`. Thiopental/propofol can be used — beware negative inotropy. Consider neuromuscular blockade if uncontrolled.
6. ARRHYTHMIAS: Wide QRS → Sodium bicarbonate `1-2mmol/kg IV` (not in AAGBI QRH).
7. INTRALIPID 20%: Bolus `1.5ml/kg IV over 2–3 min`. Immediately start infusion `15ml/kg/hour`. Document cumulative dose.
8. RE-BOLUS: At 5 and 10 min if not stable or circulation deteriorates.
9. DOUBLE INFUSION RATE: At any time after 5 min if not stable → `30ml/kg/hour`.
10. MAX DOSE: Do NOT exceed `12ml/kg` cumulative. Stop infusion once stable.

# LAST — LOCAL ANAESTHETIC SYSTEMIC TOXICITY

**Page 2: Action Flowchart — PAEDIATRIC (<16 YRS)**

- **LAST = CNS/CVS toxicity after local anaesthetic | STOP LA injection/infusion IMMEDIATELY | Benzodiazepine for seizures | Lipid is the antidote**

## 1 STOP & CALL

- Stop local anaesthetic (remember infusion pumps).
- Call 2222. Alert anaesthetics + PICU.
- High-flow O2. IV/IO access. Continuous cardiac monitoring.

## 2 TREAT SEIZURES

- Midazolam `0.1mg/kg IV/IO` (max 5mg if <20kg, max 10mg if >20kg) GIVE NOW
- Benzodiazepine = drug of choice. Thiopental/propofol: beware negative inotropy. Consider neuromuscular blockade.
- If refractory: Levetiracetam `40mg/kg IV` over 5 min.

## 3 INTRALIPID 20% — BOLUS

- Bolus: `1.5ml/kg IV over 2–3 min`.
- Immediately start infusion: `15ml/kg/hour`.
- Track cumulative dose — use dose chart on walkthrough.

## 4 INTRALIPID 20% — INFUSION & RE-BOLUS

- At 5 and 10 min: still unstable? Re-bolus `1.5ml/kg` (3 total).
- Any time after 5 min: double infusion to `30ml/kg/hour`.
- Continue until stable. MAX `12ml/kg cumulative`.

## 5 CARDIAC ARREST (IF OCCURS)

- PALS algorithm. Continue CPR.
- Wide QRS / arrhythmia → Sodium bicarb `1-2mmol/kg IV` (not in AAGBI QRH)
- **Adrenaline ≤1mcg/kg** (AAGBI QRH). Avoid vasopressin. Recovery may take >1 hour.
- Continue Intralipid despite arrest

MAX CUMULATIVE DOSE: 12ml/kg. STOP infusion once haemodynamically stable. Monitor for recurrence — ICU admission mandatory.



*Feedback/Updates: jake.turner@uhb.nhs.uk*

# SURGICAL AIRWAY — PAEDIATRIC CICO

**Page 1: Walkthrough — ALL AGES (EM Without ENT)**

⚠ DIFFICULT AIRWAY TROLLEY

**≥8 YRS: eFONA (DAS 2025 vertical incision) | <8 YRS: Surgical tracheotomy (CTM too small) | ANY AGE: Cannula = bridge only**

## EQUIPMENT — ALL TECHNIQUES

- ☐ **Scalpel Size 10**
- ☐ Curved scissors / Tuff Cut
- ☐ 2× Kelly clamps
- ☐ 14–18G cannula + 5ml syringe
- ☐ Suction + BVM + EtCO2
- ☐ Shoulder roll | Good lighting

**ETT BY AGE:**
Neonate: 3.0 uncuffed + 5Fr bougie
Infant (1m–1yr): 3.5–4.0 + 5Fr
Child (1–8yr): 4.0–4.5 + 8Fr Frova
≥8yr eFONA: **6.0 cuffed** + adult bougie
*Fallback: (Age÷4)+4 uncuffed*

Shiley: **<8yrs = 3.5 neo | 8–16yrs = 4.5 paed**

## CANNULA BRIDGE — ANY AGE (BUYS TIME)

**<1yr:** Puncture *trachea* (not CTM) at **45° caudally**. **>1yr:** CTM or trachea.
14–18G cannula + syringe. Air aspirated = intratracheal.
Advance cannula. Remove needle.
BVM via 3ml syringe barrel. 1 L/min/yr age (min 4 L/min). Watch chest rise.
Temporary — max ~15 mins. Hold cannula — kinks easily.

🩺 Subcutaneous emphysema = misplaced.

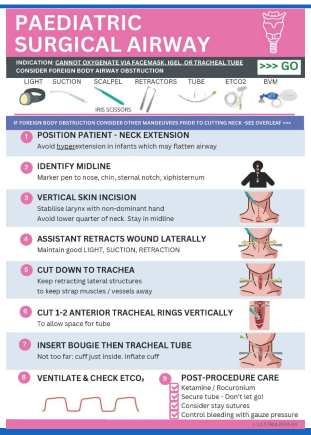
## EFONA ≥8 YEARS — DAS 2025 VERTICAL INCISION

**1.** Neck extended. Shoulder roll.
**2.** Laryngeal handshake — identify CTM.
**3. Vertical skin incision** up to 8cm, caudad → cephalad.
**4.** Blunt finger dissection → CTM. Transverse stab through CTM, rotate blade 90°.
**5.** Bougie caudally 10–15cm. Feel tracheal clicks.
**6.** Railroad **6.0 cuffed ETT**. Inflate. Confirm EtCO2.

## SURGICAL TRACHEOTOMY <8 YRS / NEONATES

**1.** Shoulder roll. Neck extended. Mark midline: nose → chin → sternal notch.
**2.** Grasp trachea — thumb + middle finger pinch. Index on midline rings.
**3. Vertical skin incision** — sternal notch upward. Anticipate bleeding.
**4.** Kelly clamps — blunt spread. Assistant retracts laterally.
**5.** Expose tracheal rings. Displace thyroid isthmus up or divide.
**6. Vertical incision rings 2–3 ONLY. NEVER horizontal** (tracheal transection risk).
**7.** Spread with clamp. Railroad Shiley 3.5 neo / ETT over bougie. Confirm EtCO2.

🩺 Neonates: rings ~1–2cm deep. Shallow + gritty. EtCO2 may be unreliable — use chest rise + USS.




Cliff Reid — Paediatric Surgical Airway v1.1 (2024) | emcrit.org | Used with permission

# SURGICAL AIRWAY — PAEDIATRIC CICO

Page 2: Action Flowchart — Age-Stratified, EM-Applicable

⚠ DIFFICULT AIRWAY TROLLEY

**⚠ DECLARE CICO — OPEN DIFFICULT AIRWAY TROLLEY — CALL 2222**
**≥8 YRS: eFONA vertical incision — 6.0 cuffed ETT (DAS 2025) | <8 YRS: Surgical tracheotomy — Shiley 3.5 neo / ETT (Age÷4)+4**
**NEONATES: Access TRACHEA (not CTM). 3.0 ETT + 5Fr bougie. Vertical only. NEVER horizontal cut.**

## 1 POSITION & PREPARE (ALL AGES)

- Shoulder roll. Neck extended. **Mark midline:** nose → chin → sternal notch.
- Grasp trachea with non-dominant hand throughout — thumb and middle finger pinch, index finger on midline.
- Announce ETT and bougie size (see walkthrough). Have cannula ready as bridge.
- 🩺 In neonates/infants — neck is short. Head may need slight chin-tilt rather than full extension. Rolls compresses airway.

## 2 CANNULA BRIDGE (ALL AGES — WHILE PREPARING)

- <1yr: Puncture **trachea below cricoid** at 45° caudal. >1yr: CTM or trachea below cricoid.
- 14–18G cannula + syringe. Air aspirated = intratracheal. Advance cannula. Remove needle. Hold throughout.
- Attach BVM via 3ml syringe barrel. Ventilate 1 L/min/yr age (min 4 L/min). Watch chest rise only.
- Bridge buys time — do NOT delay definitive airway. Cannula kinks easily. Hold it at all times.

### PATH A — eFONA (≥8 YRS)

**Step 3a:** Laryngeal handshake. Identify CTM.
**Step 4a: Vertical skin incision** (up to 8cm) → finger dissection → transverse stab through CTM, rotate 90°.
**Step 5a:** Insert bougie caudally along blade — 10–15cm. Feel clicks.
**Step 6a:** Railroad **6.0 cuffed ETT**. Inflate. Confirm EtCO2.

### PATH B — TRACHEOTOMY (<8 YRS / NEONATES)

**Step 3b:** Long **vertical skin incision** — sternal notch to chin. Spread with clamps.
**Step 4b:** Expose trachea. Displace thyroid isthmus up or divide it.
**Step 5b: Vertical incision through rings 2–3** only. NEVER horizontal.
**Step 6b:** Spread with clamp. Insert Shiley 3.5 neo / ETT over bougie. Confirm EtCO2.

## 7 CONFIRM, SECURE & ESCALATE (ALL PATHS)

- **Confirm:** EtCO2 (gold standard) + chest rise + auscultation bilaterally.
- **Secure tube** with tube tie + suture through flanges if Shiley. Mark depth at skin.
- In neonates: **EtCO2 may be unreliable** (low CO2 output) — use chest rise + lung USS as adjuncts.
- Transfer to PICU/theatre. Call ENT for formal tracheostomy in all cases.

⚠ **NO ETCO2 AFTER INSERTION**
Withdraw tube. Maintain oxygenation via cannula bridge or BVM face. Retry with fresh equipment. In neonates — if tracheal incision visible, insert 3ml syringe barrel directly into opening and attach BVM as last-resort temporising measure.

DAS / ESAIC 2024

*Feedback/Updates: jake.turner@uhb.nhs.uk*

# DC CARDIOVERSION

Page 1: Walkthrough (Preparation) — PAEDIATRIC (<16 YRS)

PAEDIATRIC EDITION

## INDICATIONS & INITIAL ACTIONS

**SYNCHRONISED (CARDIOVERSION):**
• Haemodynamically unstable SVT (HR typically >220 bpm in infants, >180 bpm in children)
• Haemodynamically unstable AF/flutter
• Ventricular tachycardia with pulse

**UNSYNCHRONISED (DEFIBRILLATION):**
• VF or pulseless VT — PALS algorithm

**STABLE:** vagal manoeuvres + adenosine first (see SVT quick reference). If they fail OR child becomes unstable → DC cardioversion.

**Call 2222:** Anaesthetics + Paediatric Reg. Sedation mandatory if conscious.

## PAEDIATRIC ENERGY & PAD GUIDE

**SYNCHRONISED CARDIOVERSION:**
• First shock: `1 J/kg` (biphasic)
• Then double each attempt: `2 J/kg` → `4 J/kg`
• Maximum: `4 J/kg` per shock

**DEFIBRILLATION (VF/pVT — unsynchronised):**
• `4 J/kg` for the initial shocks
• Refractory VF/pVT (i.e. more than five shocks are needed): increase stepwise up to `8 J/kg` (max 360 J)

**PAD SIZE:**
• **<10 kg** (approx <1yr): Paediatric paddles/pads
• **>10 kg**: Adult pads acceptable

**PAD PLACEMENT:**
• Anterior-posterior **preferred** in small children (avoids pad overlap)
• Anterior-lateral acceptable if AP not feasible

## SEDATION FOR CARDIOVERSION

**Mandatory if patient is conscious.**

**First line:** Ketamine `1–2 mg/kg` IV slow push (dissociative, maintains airway)
**Alternative:** Midazolam `0.1 mg/kg` IV + Fentanyl `1–2 mcg/kg` IV

Have: BVM + suction + intubation equipment immediately available.
Monitoring: SpO2, EtCO2 (if intubated), ECG, NIBP throughout.

In haemodynamic collapse — do NOT delay cardioversion for sedation. Cardiovert immediately, then manage pain/distress.

## PROCEDURE WALKTHROUGH

1. **Stable?** Vagal / adenosine first (SVT box below).
2. **Monitoring:** Continuous ECG, SpO2, NIBP. 12-lead ECG if time allows.
3. **IV/IO access:** Large proximal vein. IO acceptable.
4. **Sedation:** Ketamine 1–2 mg/kg IV if conscious. GIVE NOW
5. **Pads:** Apply combo pads — <10kg = paediatric; >10kg = adult. Anterior-posterior preferred.
6. **Sync Mode:** CONFIRM SYNC is ON (markers on each QRS).
7. **Charge:** 1 J/kg. Announce: *"Charging — stand clear."*
8. **Clear:** Confirm all staff away. Oxygen away from patient face.
9. **Shock:** Deliver. Assess rhythm immediately after.
10. **If no response:** Double to 2 J/kg, then 4 J/kg (max). Seek senior/cardiology input.
11. **Post-cardioversion:** 12-lead ECG. IV amiodarone if recurrent VT: `5 mg/kg` over 30 mins. Transfer to PICU.

## PAEDIATRIC SVT — QUICK REFERENCE

**ECG:** Narrow complex tachycardia, no P-waves (or retrograde P), abrupt onset/offset

**Stable SVT pathway:**
1. Vagal: ice to face <1yr | Valsalva >5yrs (straw)
2. Adenosine 0.1–0.2 mg/kg rapid IV push (max 6mg)
3. Adenosine 0.3 mg/kg if no response (max 12–18mg)
4. Synchronised DC cardioversion 1 J/kg if above fail

**Unstable SVT (shock/collapse):**
→ Synchronised DC cardioversion immediately 1 J/kg

Do NOT give adenosine to haemodynamically compromised patients — may precipitate cardiac arrest.

## EQUIPMENT LOCATION

**DEFIBRILLATOR IN RESUS BAY. PAEDIATRIC PADS IN PAEDIATRIC RESUS.**

# DC CARDIOVERSION

Page 2: Action Flowchart — PAEDIATRIC (<16 YRS)

PAEDIATRIC EDITION

- **SYNC ON for all cardioversion (non-arrest). SYNC OFF for defibrillation (VF/pVT).**
- **Pads: <10 kg = paediatric pads. >10 kg = adult pads acceptable.**
- **Sedate before shocking if conscious — Ketamine 1–2 mg/kg IV.**

## 1 STABLE SVT — TRY FIRST

- **Vagal manoeuvres:** Ice to face <1yr (10 sec) | Valsalva >5yrs (blow through straw)
- **Adenosine:** `0.1–0.2 mg/kg` IV rapid push + 10ml saline flush (max 6mg first dose). Large proximal vein. GIVE FAST
- Repeat: `0.3 mg/kg` after ≥1 min if SVT persists (max 12–18mg).
- Do NOT give adenosine if haemodynamically compromised — go straight to cardioversion.

## 2 SEDATE & PREPARE

- Ketamine `1–2 mg/kg` IV slow push (if conscious). GIVE NOW
- Apply combo pads: <10kg = paediatric; >10kg = adult. Anterior-posterior preferred.
- **CONFIRM SYNC MODE ON** — look for sync markers on QRS complexes on ECG display.

## 3 SHOCK — SYNCHRONISED CARDIOVERSION

- Set energy: `1 J/kg` (biphasic).
- Charge. Announce: *"Stand clear — oxygen away."*
- Confirm all clear. Deliver shock.
- Assess rhythm immediately. If no response → double to `2 J/kg`, then `4 J/kg` (max).
- 🩺 After each shock — reconfirm SYNC mode is still ON (some monitors reset to async after shock).

## 4 DEFIBRILLATION — VF / PVT (UNSYNCHRONISED)

- **Confirm SYNC MODE OFF.** VF/pVT = unsynchronised shock.
- Energy: `4 J/kg` for the initial shocks.
- Immediately resume CPR after shock. 2 min cycle before rhythm check.
- Refractory VF/pVT (i.e. more than five shocks are needed): increase the energy stepwise, up to 8 J/kg (max 360 J).

## 5 POST-CARDIOVERSION

- 12-lead ECG immediately.
- Recurrent/refractory VT: Amiodarone `5 mg/kg` IV over 30 mins. FETCH DRUG
- Recurrent SVT: Discuss with Paediatric Cardiology — consider adenosine repeat or digoxin.
- Transfer to PICU. Continuous cardiac monitoring.

⚠️ **NO RESPONSE AFTER 3 SHOCKS**

Call Paediatric Cardiology urgently. Consider: correct paddle position, adequate contact, sync mode status, underlying reversible cause (electrolytes, hypoxia). Magnesium `25–50 mg/kg` IV (max 2g) for torsades de pointes.



*Feedback/Updates: jake.turner@uhb.nhs.uk*

Reference: Resuscitation Council UK Paediatric Life Support 2025 (incl. paediatric arrhythmia algorithm) | Advanced Paediatric Life Support 7th Ed. (2023)

Open RCUK PALS Guidelines

# WETFLAG — PAEDIATRIC EMERGENCY REFERENCE

Weight · Energy · Tube · Fluids · Lorazepam · Adrenaline · Glucose

RCUK PALS 2025

**WEIGHT ESTIMATION (if scales unavailable) | Use Broselow tape where possible**
<1 yr: (0.5 × age in months) + 4 | 1–5 yr: (2 × age) + 8 | 6–12 yr: (3 × age) + 7

| Letter | Parameter | Formula / Dose | Clinical Notes |
|---|---|---|---|
| W | **Weight** (kg) | <1yr: (0.5×months)+4<br>1–5yr: (2×age)+8<br>6–12yr: (3×age)+7 | Foundation for all calculations. Use measured weight or Broselow tape where available. All doses below are based on estimated weight. |
| E | **Energy** (defibrillation) | 4 J/kg<br>Initial shocks<br>Refractory VF/pVT (i.e. more than five shocks are needed): escalate stepwise to 8 J/kg (max 360 J) | VF / pulseless VT only. Synchronised cardioversion (SVT): 1 J/kg first, doubling to max 4 J/kg. Max 360 J (RCUK 2025). |
| T | **Tube (ETT)** (size & depth) | Uncuffed ID: (Age÷4)+4<br>Cuffed ID: (Age÷4)+3.5<br>Oral depth: (Age÷2)+12 cm<br>Nasal depth: (Age÷2)+15 cm | Neonates: 3.0–3.5 uncuffed. Confirm position clinically + EtCO2. X-ray if time allows. Oral depth mark at lips. Cuffed tube ID is 0.5 smaller than equivalent uncuffed. |
| F | **Fluids** (resuscitation) | Medical: 10 ml/kg<br>Trauma: 10 ml/kg | Hartmann's / Plasmalyte (balanced crystalloid). Reassess after each bolus; repeat up to 40–60 ml/kg in first hour if needed (RCUK 2025). Trauma: limit crystalloid to 20ml/kg max — use blood products early. |
| L | **Lorazepam** (seizure) | 0.1 mg/kg<br>Max single dose: 4 mg | IV or IO. Buccal midazolam 0.3mg/kg (max 10mg) if no IV access. If seizure continues after 2 doses → call anaesthetics, consider levetiracetam 40mg/kg IV (over 5 min) or phenytoin 20mg/kg IV. |
| A | **Adrenaline** (cardiac arrest) | 0.1 ml/kg of 1:10,000<br>= 10 mcg/kg (max 1 mg).<br>Repeat every 3–5 mins. | IV or IO. Give as early as possible in non-shockable arrest (PEA/asystole). In shockable arrest: give after 3rd shock, then every alternate cycle. Do NOT give via ETT. |
| G | **Glucose** (hypoglycaemia) | 2 ml/kg of 10% dextrose | IV or IO bolus. Recheck BM after 5–10 mins; repeat if needed (RCUK 2025). If no 10%: use 5ml/kg 5% dextrose. Never give 50% dextrose undiluted to children — causes vein damage. |

### AMIODARONE (SHOCKABLE ARREST)

5 mg/kg IV/IO after 3rd shock (max 300mg).
Repeat 5 mg/kg after 5th shock (max 150mg). Do NOT delay defibrillation.

### ATROPINE (BRADYCARDIA)

20 mcg/kg IV/IO (max 500mcg up to 11 yrs; 300–600mcg at 12–17 yrs — RCUK 2025).
Only for vagally-mediated or conduction disease bradycardia.

### IO ACCESS (IF NO IV IN 5 MINS)

Proximal tibia: 1–3cm below and medial to tibial tuberosity (~1cm neonates, ~2cm children).
Distal femur: midline, 2–3cm above lateral condyle.
Humeral head: arm on abdomen, shoulder internally rotated.

*Feedback/Updates: jake.turner@uhb.nhs.uk*

# STOP 5 HOT DEBRIEF

Perform immediately after the event, before the team disperses. 5 minutes maximum.

1. **S**ummarise the case (1 minute).
2. **T**hings that went well?
3. **O**pportunities to improve?
4. **P**oints for action (Fix equipment, report incidents).
5. **5** minutes max — then check on each other.

🩺 **Paediatric-Specific Debrief Questions:**
- Was age-appropriate equipment used (ETT size, pacing pads, drain size)?
- Was weight estimated correctly (Broselow tape / formula)?
- Were age-appropriate drug doses used (weight-based)?
- Was IO access available and used appropriately?
- Was the correct surgical airway technique used for the patient's age (eFONA ≥8 yrs / Tracheotomy <8 yrs)?
- Was the Paediatric Registrar involved?
- Was DATIX / incident report completed?

Is everyone okay? Does anyone need support right now?
*Check on your colleagues — high acuity paediatric events are extremely distressing.*

Escalate to Consultant within 24 hours. Complete team debrief within 48 hours.